The goal of this project is to create an exploration and mapping robot named GiroBot. This system is designed to meet the needs of two main types of users. First, it can be used by the military as a guide for reconnaissance missions and topographic surveys on rough terrains. Second, it is highly useful for architects and the construction industry to map the inside of buildings and to help organize spaces. The main problem this robot solves is the ability to easily access and map areas that are usually hard to reach.

The user will pilot the robot with a remote and will not have to worry about keeping it balanced. The interaction with the system works through two different interfaces. The pilot uses a wireless remote control to manage the movements of the robot, make it jump over obstacles, and check essential data. At the same time, a computer interface will receive the data wirelessly. This computer software will process the images and display the detailed maps created by the onboard camera and the Lidar sensor.

To make all of this possible, the GiroBot must operate under specific technical constraints. The robot moves on only two wheels which makes it more maneuverable in small and tight places. To stay perfectly stable on its two contact points, it uses a gyroscope and carefully manages its own weight. The robot is also designed to drive on different types of outdoor surfaces like grass and dirt. The system must be compact with a height of about thirty centimeters and it needs to have a battery life of at least forty minutes to complete its missions.